Astra NX-200 Industrial Edge Gateway - Hardware Datasheet (Page 1)

Part Number: AST-NX200-R2 | Hardware Revision: Rev 2.4

1. Product Summary and Compute Architecture

The Astra NX-200 (model AST-NX200-R2) is a rugged, fanless DIN-rail industrial edge gateway engineered for deterministic factory telemetry, protocol translation, and air-gapped plant-floor data aggregation.

Processor: Quad-core ARM Cortex-A78AE industrial SoC operating at 2.2 GHz.
System Memory: 8 GB LPDDR5 with inline Error-Correcting Code (ECC) at 4800 MT/s.
Primary Storage: 64 GB industrial-grade pSLC eMMC 5.1 flash soldered on board.
Expansion Storage: One internal M.2 2280 Key-M PCIe Gen3 x2 NVMe slot supporting up to 2 TB industrial NVMe SSD modules.
Security Element: Dedicated discrete Infineon OPTIGA TPM 2.0 chip supporting hardware secure boot and cryptographic key storage.

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2. Electrical Power Specifications

Input Voltage Range: Dual redundant terminal-block DC inputs rated 12 V to 48 V DC.
Power Consumption: Typical nominal power draw is 18 W; peak load with PoE+ active is 34 W.
Protection Circuitry: Integrated reverse-polarity protection up to -60 V DC and over-voltage crowbar clamping above 54 V DC.

3. Mechanical and Environmental Ratings

Operating Temperature: -30 C to +70 C continuous ambient with vertical DIN-rail airflow.
Storage Temperature: -40 C to +85 C non-operating.
Relative Humidity: 5% to 95% non-condensing.
Enclosure Protection: IP40 black anodized extruded aluminum chassis, weight 1.15 kg.
Vibration Resistance: 5 Grms random vibration (5 Hz to 500 Hz) per IEC 60068-2-64.
Shock Resistance: 50 G half-sine 11 ms pulse per IEC 60068-2-27.

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4. Ethernet and Industrial Fieldbus Interfaces

Ethernet Port ETH0: 2.5GbE RJ45 uplink/management port (10/100/1000/2500BASE-T).
Ethernet Port ETH1: 2.5GbE RJ45 downstream sensor port featuring IEEE 802.3at PoE+ Power Sourcing Equipment (PSE) delivering up to 25.5 W (requires >= 24 V DC input).

Serial Ports COM1 & COM2: Two software-selectable RS-485 / RS-232 DB9 ports featuring 2.5 kVrms digital galvanic isolation and programmable 120-ohm line termination.
Native protocol support for Modbus RTU master/slave up to 921.6 kbps.

CAN Bus Interface: One isolated CAN-FD phoenix terminal port supporting 1 Mbps nominal arbitration bit rate and up to 5 Mbps data-phase bit rate (ISO 11898-1:2015).
Digital I/O: 4 optically isolated digital inputs (24 V wet/dry) and 2 Form-A relay outputs.